



Atomic Layer Deposition for Coating of High Aspect Ratio TiO₂ Nanotube Layers

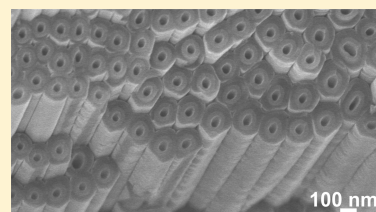
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ABSTRACT: We present an optimized approach for the deposition of Al₂O₃ (as a model secondary material) coating into high aspect ratio (≈ 180) anodic TiO₂ nanotube layers using the atomic layer deposition (ALD) process. In order to study the influence of the diffusion of the Al₂O₃ precursors on the resulting coating thickness, ALD processes with different exposure times (i.e., 0.5, 2, 5, and 10 s) of the trimethylaluminum (TMA) precursor were performed. Uniform coating of the nanotube interiors was achieved with longer exposure times (5 and 10 s), as verified by detailed scanning electron microscopy analysis. Quartz crystal microbalance measurements were used to monitor the deposition process and its particular features due to the tube diameter gradient. Finally, theoretical calculations were performed to calculate the minimum precursor exposure time to attain uniform coating. Theoretical values on the diffusion regime matched with the experimental results and helped to obtain valuable information for further optimization of ALD coating processes. The presented approach provides a straightforward solution toward the development of many novel devices, based on a high surface area interface between TiO₂ nanotubes and a secondary material (such as Al₂O₃).



INTRODUCTION

Over the past 20 years, self-organized valve metal oxide nanoporous or nanotubular structures have attracted huge scientific and technological attention, due to their unique architecture and intriguing properties. In particular, this accounts for nanoporous anodic alumina^{1,2} and nanotubular anodic titania^{3–6} prepared by a low-cost electrochemical anodization of corresponding metal substrates. Nanoporous alumina, typically in the form of membranes, has been mainly employed as the templating or supporting material for synthesis of various functional materials and devices.^{7–12} In contrast to nanoporous alumina, TiO₂ nanotube layers have been exploited for a significantly larger number of applications, owing to the semiconductive nature of TiO₂, unique tubular architecture, and chemical stability. Outstanding performance of TiO₂ nanotubes was revealed mainly in photocatalysis, solar cells, self-cleaning, and biomedical fields^{13–15} among others. Significant and valuable efforts were carried out to tune the aspect ratio of the nanotubes,^{5,6,16–18} to improve the tube ordering,^{19–24} crystallinity,^{25–29} and to prepare tube layers on various substrates, including conductive glasses^{30–32} for various functional devices. On the other hand, comparably smaller efforts were devoted to obtain a uniform coating of the tubes with a secondary material, such as metals, oxides (including those with semiconducting properties), quantum dot materials, conducting polymers, or chalcogenides. Until now, numerous deposition approaches were reported to coat or fill the interior

parts of the nanotubes, including electrodeposition,^{33–36} chemical bath deposition,^{37–40} spin-coating,^{41,42} sputtering,^{43–45} and atomic layer deposition (ALD). While ALD is one of the most promising deposition techniques for its excellent homogeneity and thickness accuracy, there have been only few reports published employing this technique for an introduction of the secondary material in the nanotube layers.^{46–53} In particular, ALD has recently been reported for the deposition of Al₂O₃, as a secondary material, onto TiO₂ nanotubular structures for enhanced water splitting⁵² and more efficient dye sensitized solar cells.⁵³ The Al₂O₃ coating resulted in an improvement of the electrochemical and photovoltaic performance displayed, both ascribed to the passivation of the surface states that leads to a reduction of the electron–hole recombination rate at the surface of the TiO₂ nanotube layer electrode. On the other hand, these two reports^{52,53} did not provide any information about the influence of ALD parameters (especially precursor diffusion times) on the overall coating nor did it contain any quartz crystal microbalance data. Nevertheless, these publications confirm that an introduction of the secondary material in the nanotubes, by a very uniform and precisely controllable deposition process, results in many advanced functionalities of the newly prepared composite

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TiO₂-based nanotube layers, similarly as it did for nanoporous materials.^{54,55} This can be especially true for high aspect ratio TiO₂ nanotube layers that are even more promising for applications than their lower aspect ratio counterparts.^{13–15} However, coating of high aspect ratio nanostructures is relatively time-consuming and demanding in terms of precursor doses.^{54,56} Thus, to avoid unnecessarily long processes and consumption of expensive precursor(s), the optimization of the ALD coating of high aspect ratio TiO₂ nanotube layers with secondary materials is highly demanded. It was only recently shown by Macak et al. that nanotube layers with an aspect ratio of ~ 80 can obtain a uniform In₂O₃ coating along the interiors of TiO₂ nanotube layers resulting in significantly enhanced antireflection performance.⁵¹ However, the diffusion of the precursors was far from being optimized and the resulting coating had unequal thickness along the tube walls. Nevertheless, this paper indicated how crucial it is to enable within the ALD process the proper diffusion of the precursors inside the nanotubes to achieve the same coating thickness throughout the whole nanotube layer.

Thus, in the present work, a detailed study on the deposition process to achieve a uniform coating of very high aspect ratio TiO₂ nanotube layers (≈ 180) with a secondary material by ALD is reported for the first time including detailed SEM analyses of the coatings. The aspect ratio in this work is considered as the ratio between the nanotube layer thickness (20 μm) and the average tube diameter (110 nm) at the top of the nanotube layer. Thus, it accounts for approximately 180. As a model secondary material, aluminum oxide (Al₂O₃) was deposited from trimethyl-aluminum (TMA) and water precursors. To study the influence of the TMA diffusion time on the thickness of the Al₂O₃ coating along the walls of TiO₂ nanotubes, different TMA exposure times during the process were employed. A thorough scanning electron microscopy (SEM) analysis was employed to evaluate the coating thickness of the interior nanotube surface at different depth levels inside the nanotube layer: top, near top, center, and bottom. In addition, quartz crystal microbalances (QCMs) with attached TNT membranes were employed to monitor the ALD process in the TNTs. This novel approach enables essentially a more detailed understanding of the applied coating process and provides new insights into the demands and challenges of ALD coating of high aspect ratio nanostructures. Finally, the theoretical minimum TMA precursor exposure time was calculated in order to get a deeper insight on the deposition process of a secondary material inside the tubes.

■ EXPERIMENTAL SECTION

The TiO₂ nanotube layers, with a thickness of $\sim 20\ \mu\text{m}$ and a nanotube diameter of $\sim 110\ \text{nm}$ ($\text{AR} \approx 180$), were prepared by anodization of Ti foils (Sigma-Aldrich, 0.127 mm thick, 99.7% purity) at 60 V for 4 h (the sweep rate was 1 V/s). Prior to the anodization, the Ti foils were degreased by sonication in isopropanol and acetone and then rinsed with isopropanol and dried in air. The anodization itself was carried out at room temperature in an ethylene glycol electrolyte containing 170 mM NH₄F (both Sigma-Aldrich, reagent grade) and 1.5 vol % deionized water. Before the first use, the electrolyte was aged for 9 h (for details, see ref 57). The electrochemical cell consisted of a high-voltage potentiostat (PGU-200V, Elektroniklabor GmbH) in a two-electrode configuration, with a Pt foil as the counter electrode and a Ti foil as the working electrode. After anodization, the Ti foils were rinsed and sonicated in isopropanol and dried in air.

The nanotube Al₂O₃ coating was achieved using an ALD process carried out in a cross-flow process chamber manufactured by FHR

Anlagenbau. Trimethylaluminum (TMA, STREM, 98% purity) and deionized water have been evaporated and delivered by bubbling 50 sccm argon carrier gas through stainless steel bubblers at 16 and 35 °C bubbler temperatures, respectively. The process chamber temperature was set to 200 °C, while pressure was kept at 50 Pa, by controlling pumping power by a butterfly valve. The ALD process consisted of 200 cycles at a deposition temperature of 200 °C, with TMA and H₂O as precursors in alternating pulses. Argon was used as carrier and purging gas. The nominal Al₂O₃ coating thickness was 27 nm. The application of different exposure times of 0.5, 2, 5, and 10 s for the Al precursor (TMA) allowed us to evaluate the influence of the TMA precursor diffusion time on the degree of coating of the tube's interior. Due to the higher viscosity of the oxidizing precursor (H₂O), the exposure time was set for 10 s for all processes and it was long enough to attain the saturation of the whole nanotube layer. Purging times for both precursors were set up long enough (20 s for all processes) to ensure their proper elimination from the ALD chamber and to avoid any undesirable gas reaction between the precursors.

The QCM measurements were carried out using an Inficon SQM-160 QCM controller and standard 6 MHz AT-cut quartz crystals. Identical TiO₂ nanotube layers (dimensions, aspect ratio) to those subjected to main ALD runs were employed as QCM detector substrates. To prepare stable and robust nanotube-based QCM crystals, free-standing nanotube layers had to be obtained first by dissolution of Ti substrate using Br₂–MeOH solution.⁵⁸ In the next step, the layers were quantitatively transferred and attached on the conventional QCM crystal (14 mm diameter, gold coated, CNT06RCIG, Colnatec) using a small amount of polymeric binder (ethanolic solution containing 9 wt % polyvinylpyrrolidone, 2 μL per crystal).

The structural characterization of the TiO₂ nanotube layers before and after ALD runs was carried out by a field-emission scanning electron microscope (FE-SEM JEOL JSM 7500F) and a scanning transmission electron microscope (STEM, FEI Tecnai F20 X-Twin) fitted with a high angle annular dark field (HAADF) detector and operating at 200 kV. The cross-sectional views were obtained from mechanically bent samples. Due to the rupture of the nanotube layers by this bending, it was possible to visualize nanotubes within the layers and coatings within nanotubes in various directions and nanotube layer depths. These visualizations allowed detailed analyses and measurements of nanotube wall thicknesses and inner nanotube diameters from the very top to the very bottom. Dimensions of the inner diameters and wall thicknesses of the nanotubes were measured and statistically analyzed by NanoMeasure software. Average values and standard deviations were calculated for all the measurements performed. It turned out from these analyses that the nanotube inner diameter, nanotube wall thickness, and nanotube layer depth are strictly related. Hence, the nanotube depth (relevant for the measurement of the Al₂O₃ coating thickness) can be determined from either nanotube wall thickness or inner nanotube diameter values.

■ RESULTS AND DISCUSSION

High aspect ratio (≈ 180) TiO₂ nanotube layers were fabricated by anodic oxidation of Ti substrates (for details, see the [Experimental Section](#)). The thickness of the nanotube layer was $\sim 20\ \mu\text{m}$, and the inner nanotube diameter at the top of the nanotube layer was $\sim 110\ \text{nm}$. [Figure 1a](#) shows a cross-sectional image of the TiO₂ nanotube layer obtained from mechanically bent TiO₂ nanotube layers introducing the four nanotube depth levels: top, near top, center, and bottom. Due to the rupture of the nanotube layer (upon the layer bending), it was possible to image individual nanotubes (and measure their dimensions) at different depths, provided that the absolute depth (from the top of the layer) was very precisely monitored, as described in the [Experimental Section](#). [Figure 1b](#) provides a schematic cross section of the nanotube structure, with a gradient in the nanotube wall thickness. Due to this gradient,

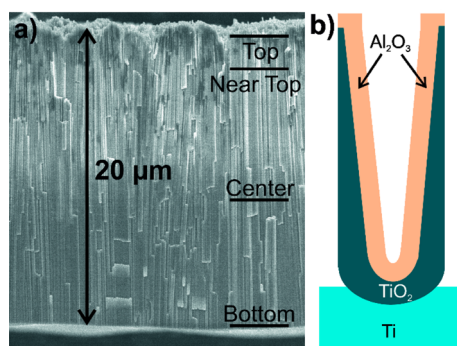


Figure 1. (a) Cross-sectional SEM image of the 20 μm thick TiO_2 nanotube layer with four different depth levels introduced in this work. (b) Cross-sectional profile of the nanotubes showing a gradient in the inner tube diameter and the Al_2O_3 coating of the tube interiors.

the inner diameter tube decreases from the top to the bottom of the nanotube layer so there is a corresponding tube diameter gradient. The unique capabilities of ALD for a uniform deposition of a secondary material into nanotubes were utilized for the nanotube coating with Al_2O_3 using trimethyl-aluminum (TMA) and H_2O precursors (for further information, see the [Experimental Section](#)). In order to evaluate the influence of the TMA precursor diffusion time on the degree of coating of the tube's interior, different TMA exposure times of 0.5, 2, 5, and 10 s were applied. The diffusion time of the oxidizing H_2O precursor and purging times for both precursors were kept constant for 10 and 20 s, respectively, throughout the whole work.

Figure 2 shows SEM images of the TiO_2 nanotube layer coated by Al_2O_3 using ALD and 5 s TMA exposure time. **Figure 2a–d** shows Al_2O_3 -coated TiO_2 nanotubes at different depth levels of the nanotube layers indicated in **Figure 1**. Further evidence of Al_2O_3 continuous coatings within nanotubes is given in **Figure 2e**, which shows Al_2O_3 coatings protruding out of TiO_2 nanotubes cracked across their wall and in **Figure 2f** which shows Al_2O_3 coatings embedded within TiO_2 nanotubes cracked along their walls. From all of these images, it is evident that the Al_2O_3 coating was homogeneous, pinhole-free, and conformal all along the nanotubes.

In addition, analysis of tube layers confirmed the complete coating of the nanotube interiors by Al_2O_3 for all used exposure times. On the basis of SEM images, Al_2O_3 coating thicknesses, TiO_2 nanotube wall thicknesses, and TiO_2 nanotube diameters were measured and statistically analyzed by NanoMeasure software. Average values and standard deviations were calculated for all the measurements performed. The nanotube diameter and wall thickness are directly linked with the depth of the nanotube layers. Thus, it was possible to obtain the dependence of both values on the depth of the nanotube layers that was with an advantage used in the identification of the actual depth, used in **Figure 2**.

On the basis of the SEM visualization and the performed statistical evaluation, two main features can be observed in **Figure 2**. First, the characteristic nanotube V-shape shown in **Figure 1b** is clearly reflected in the progressive tube inner diameter, narrowing with an increasing depth of the nanotubes. Second, it is noteworthy that a decreasing thickness of the Al_2O_3 coating was observed at the deepest parts in all the cases studied.

Figure 3a shows the Al_2O_3 coating thickness measured for all the exposure times as a function of both the inner nanotube

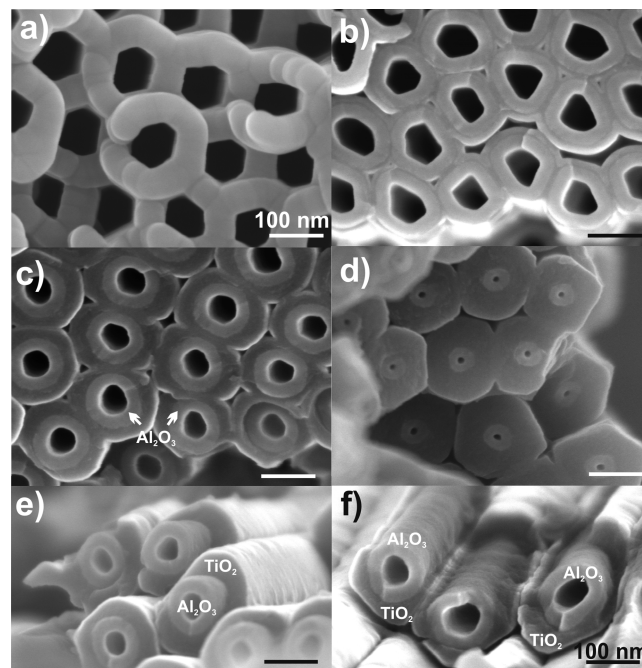


Figure 2. SEM images of TiO_2 nanotubes coated by Al_2O_3 using ALD with 5 s TMA exposure time. Images taken at four depth levels: (a) top, (b) near top, (c) center, and (d) bottom. Further evidence of Al_2O_3 continuous coatings within nanotubes: (e) Al_2O_3 coatings protruding out of nanotubes cracked across their wall and (f) Al_2O_3 coatings embedded within nanotubes cracked along their walls. The scale bar represents a distance of 100 nm.

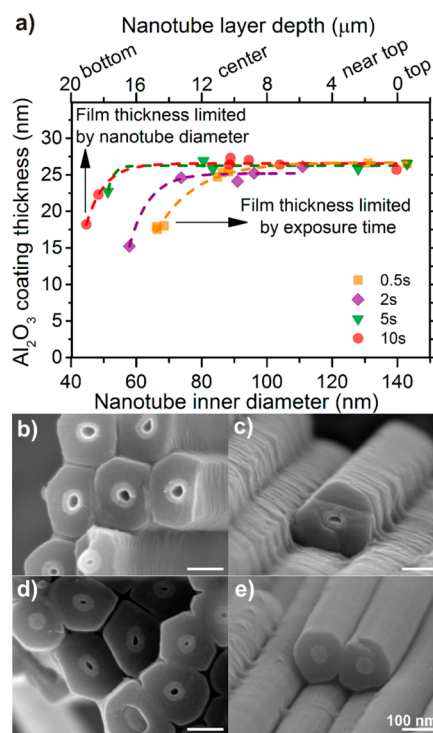


Figure 3. (a) Thickness of Al_2O_3 coating (by ALD) as a function of the inner TiO_2 tube diameter and tube depth (0 μm stands for the top part of the tube layer and 20 μm for the bottom part). SEM images taken from nanotube layers for the different TMA exposure times at the tube bottom parts: (b) 0.5 s, (c) 2 s, (d) 5 s, and (e) 10 s. The scale bar represents a distance of 100 nm.

diameter and the nanotube depth, as described in the [Experimental Section](#). Therein, it can be observed how the coating thickness decreased from a nominal value of ~ 27 nm to lower values at the deepest parts of the nanotubes. In the case of samples with short exposure times (0.5 and 2 s), it is ascribed to an insufficient exposure time of the precursors to coat the deepest levels of the tubes. In principle, in a deposition process beyond the diffusion boundary, a drop of the coating thickness would be noticeable, as it is clearly perceived in [Figure 3b,c](#) for short exposure times (0.5 and 2 s). In contrast, the Al_2O_3 coating thickness for long exposure time (5 and 10 s) at the deepest levels of tubes decreases, as it is physically limited within a narrowing inner tube diameter, as a consequence of the tube wall thickness gradient (as shown in [Figure 1b](#)). In other words, the coating cannot grow thicker, as there is no available space for it due to the narrow inner tube diameter. Therefore, the reduction in the Al_2O_3 coating thickness stems from morphological limitations, and not from an insufficient precursor diffusion time, as shown in [Figure 3d,e](#), where the Al_2O_3 coating nearly fills the tube interiors, leaving a hole with a diameter of a few nanometers in the coating. Overall, the analysis of the SEM images confirmed the successful homogeneous and conformal coating process of the interior tube surface by a secondary material (Al_2O_3).

In situ QCM measurements were carried out to monitor the ALD process, in particular the variation of the mass increment per cycle. Identical TiO_2 nanotube layers to those layers subjected to main ALD deposition runs (shown in [Figures 2](#) and [3](#)) were employed as QCM detector substrates. This was done in order to ensure the full compatibility and compliance of the results obtained from QCM with the results of ALD runs. To carry out QCM measurements without the damage or loss of the nanotube layer, stable and robust nanotube-based QCM crystals were prepared using a tailored route (see the [Experimental Section](#) for details).

[Figure 4a](#) shows the QCM results (expressed as the frequency change) from two sets of consecutive ALD cycles registered during two different stages of the Al_2O_3 coating process. The QCM staircase resulted from the alternating TMA and H_2O pulses injected into the deposition chamber during the Al_2O_3 coating process, which took place in discrete steps. Parts b and c of [Figure 4](#) show essentially the same data but expressed as the mass increase and cumulative mass, respectively, after recalculation using Sauerbrey's equation with standard quartz crystal properties.⁵⁹ From these plots, it is clear that the QCM measurements displayed the frequency of the mass variation on the sample after each precursor exposure. As is particular evident from [Figure 4b](#) and c, the major mass gain occurred during the TMA exposure time followed by a slight mass gain after the H_2O exposure period. The slight decrease in mass after the H_2O exposure time has been accounted for recombination of surface hydroxyl groups or desorption of molecular water.⁶⁰

QCM measurements in [Figure 4b](#) and c show a higher mass uptake during the first stage of the coating process (cycles 11–15) than for a posterior stage (cycles 75–79). The decrease of the deposited mass can be assigned to the reduction of the available surface within the nanotube layer. This is further clear from [Figure 4d](#) that shows a normalized comparison of the TMA saturation between ALD cycles 11 and 75. It shows a faster TMA surface saturation at cycle 75 than at cycle 11, and that is a clear proof of reduced surface area. This surface reduction can be described by the following physical scenario.

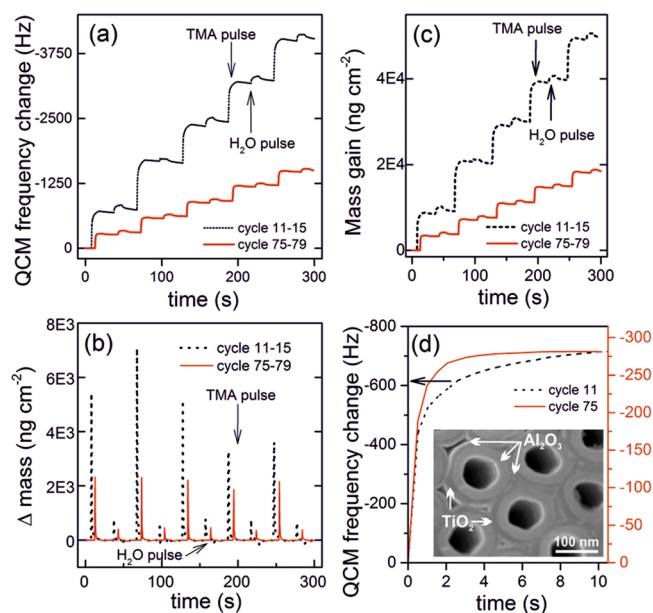


Figure 4. QCM measurements showing the frequency change per cycle as a function of the deposition time, recorded during Al_2O_3 coating of the TiO_2 nanotube layer for 10 s TMA exposure time, expressed as (a) the frequency change, (b) the mass increase, and (c) cumulative mass uptake that corresponds to the TMA and H_2O exposures (as indicated by arrows), and (d) normalized comparison of TMA saturation curves. For comparison, two different stages of the coating process (corresponding to a specific number of cycles) are shown here. The inset in part d shows a SEM image revealing the clogging of the intertube space by the Al_2O_3 coating.

At the initial cycles of the ALD process, the Al_2O_3 coating was deposited both on the exterior and interior TiO_2 nanotube surface. However, a progressive clogging of the intertube space occurred with increasing number of cycles due to the growth of the Al_2O_3 coating, which gradually hampered the diffusion of the precursor molecules toward the exterior surface of the tubes. Once the Al_2O_3 coating clogged the intertube space, the deposition processes continued only on the top and within the interior TiO_2 nanotube surfaces but no longer on the exterior surface. Such reduction of the available surface for Al_2O_3 coating was reflected on a lower uptake mass per cycle (cycles 75–59). The clogged intertube space regions by the deposited Al_2O_3 coating were clearly distinguishable, as demonstrated in the SEM inset in [Figure 4d](#), where also the Al_2O_3 coating was discerned onto the exterior and interior surfaces of the nanotubes. The Al_2O_3 coating on the external nanotube surface was very thin. It could not grow thicker due to the clogging of the intertube space. On the basis of the QCM results shown in [Figure 4](#), it also has to be pointed out that exposure times longer than 10 s did not have any beneficial effect on the coating of the nanotubes and only led to unnecessary prolongation of the ALD process time.

All of these results verified the scenario describing the decreasing mass uptake into the TiO_2 nanotube layer, resulting from the reduction of its available surface, caused by the clogging of the intertube space. After the intertube clogging had been reached, the frequency per cycle reached a quasi-plateau (cycles 75–79 shown in [Figure 4a](#)) related to an exclusive Al_2O_3 coating on the top, and within the interiors of the nanotubes. Using this quasi-plateau process, parameter variations have been carried out to determine the required

TMA and H₂O doses, as well as the impact of process temperature and pressure (data not shown here). Overall, these QCM measurements were insightful for the characterization of the Al₂O₃ coating processes carried out within the nanotube layers by ALD.

In addition, theoretical calculations were performed (i) to determine the minimum exposure time to achieve the uniform deposition in the high aspect ratio nanotubes and (ii) to get a deeper understanding on the Al₂O₃ coating process by ALD. We based the theoretical calculations on the work of Elam et al.⁵⁶ who used nanoporous anodic alumina membranes as an ideal template for exploring a thin film deposition by ALD into a high aspect ratio structures. They obtained, by Monte Carlo simulations, a mathematical expression that describes the complete and conformal coverage on high aspect ratio nanostructures in the diffusion-limited regime, and provides the minimum exposure time required for such. The mathematical expression described by Elam et al. is as follows

$$t = 2.3 \times 10^{-7} P^{-1} (m)^{1/2} \Gamma (L/d)^2 \quad (1)$$

where t is the time in seconds, P is the reactant pressure in Torr, m is the mass of the reactant molecule (TMA) in amu (72), Γ is the density of ALD reactive sites in 10^{15} cm^{-2} (0.2391), d (11 nm) is the tube diameter, and L (20 μm) is the nanotube length. The density of reactive sites was calculated from the density of the coating (3.0 g cm^{-3}) and the ALD growth rate. The calculation of the minimum exposure criterion should be estimated from the final aspect ratio of the nanotubes following the complete deposition. In our case, due to the characteristic tube interior diameter gradient, a nanotube layer depth with a diameter of 65 nm was considered, which after a nominal deposition of 27 nm leads to a final tube diameter of 11 nm. The selection of a nanotube layer depth level appropriate to this diameter was not trivial. It had to fulfill two main conditions. First, the selected nanotube layer depth must not have clogged so that the precursor diffusion could continue during the whole deposition process. Second, the selected nanotube layer depth should be as deep as possible to preserve the high aspect ratio characteristic of the structure. Nanotube layer depths deeper than the selected one did not satisfy the first condition, as they clogged before the deposition process was completed. No precursor diffusion could take place there. On the other hand, shallower nanotube layer depths did not fulfill the second condition.

Due to the configuration of the ALD facility, it was not possible to determine the TMA partial pressure value required for the theoretical calculation of the minimum exposure time to obtain uniform coating in the high aspect ratio nanotubes. The precursor TMA was injected into the deposition chamber together with Ar carrier gas, and the total pressure value was known to be 50 Pa. Thus, eq 1, considering the exposure time as a variable, allowed us the advantage of calculating the TMA partial pressure for different exposure times. TMA partial pressure values for exposure times of 2, 3, 4, and 5 s were calculated. It was revealed that the TMA partial pressure values (just below 50 Pa) corresponded to exposure times between 4 and 5 s. Shorter exposure times were ruled out, as they led to higher TMA partial pressure than the total set pressure (50 Pa). Thus, the calculated exposure times matched coherently with the experimental data shown and discussed in Figures 3 and 4.

The predicted minimum exposure times are only valid in the diffusion-limited regime, i.e., in the limit that $S \gg H$, where the reactive sticking coefficient, S , is much greater than the hopping

coefficient, H . The hopping coefficient value is a function of the aspect ratio of the structure, $H = 16(d/L)^2$, while the reactive sticking coefficient⁶⁰ for Al₂O₃ coating by ALD is $S = 1 \times 10^{-3}$. The corresponding hopping coefficient H is 4.84×10^{-6} , and therefore, the condition of $S \gg H$ is fulfilled. Thus, the predicted minimum exposure time can be considered valid and the diffusion-limited regime can be identified for the presented ALD process in this work.

Finally, in order to assess the quality of the Al₂O₃ coating, scanning transmission electron microscope (STEM) analyses were carried out. Highly uniform, continuous, and pinhole-free Al₂O₃ coatings on the inner wall of the TiO₂ nanotube were confirmed by STEM imaging of individual nanotubes, as shown in Figure 5 which shows a fragment of the nanotube from the

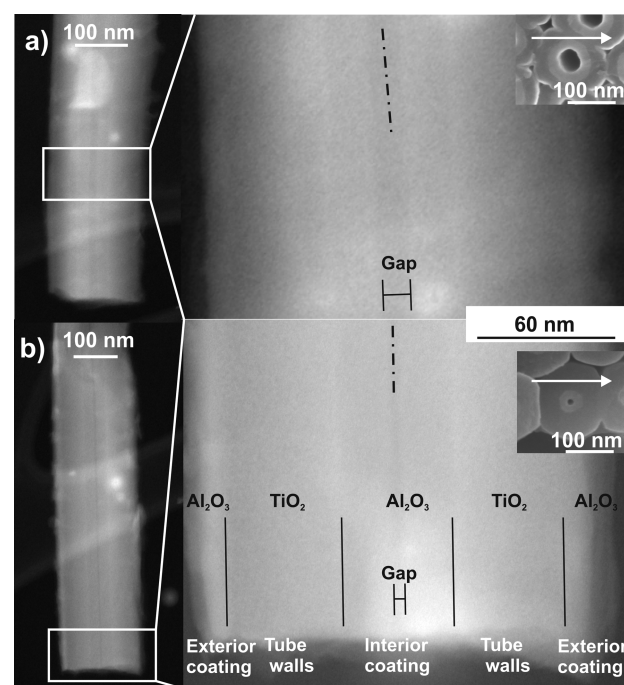


Figure 5. STEM-HAADF images of fragments of Al₂O₃ coated TiO₂ nanotubes taken from the (a) the upper part and (b) the bottom part of the TiO₂ nanotube layer. The dash-dot lines exhibit the geometrical center axis of the nanotubes. Interfaces between individual parts of the tubes are distinguished by solid lines and appropriate description. Insets show SEM images of corresponding tube parts, where arrows indicate the direction of the STEM imaging through the whole tube.

upper part (Figure 5a) and from the bottom part (Figure 5b) of the TiO₂ nanotube layer. In particular, the high magnification STEM images of the nanotubes clearly reveal the TiO₂ walls, continuous Al₂O₃ coating on both the inner and outer TiO₂ walls, and the gap (pore) in the tube center that can also be seen from SEM images shown as insets (and as essentially demonstrated in Figure 2).

All in all, from all presented results, it is clear that ALD with optimized diffusion of precursors can be used for the preparation of a range of coatings with different thicknesses and diverse compositions. From this point of view, Al₂O₃ should only be considered as a model material. Many more different materials, such as other oxides, sulfides, nitrides, carbides, etc., can be deposited inside the TiO₂ nanotubes (or virtually any high aspect ratio porous nanostructure) by means of ALD, yielding new functionalities. The results presented here

serve as the proof-of-principle that ALD is a completely viable tool to prepare uniform coatings within nanotube layers with a thickness, that can be driven by the ALD operator and that stems from the desired application of the coating. It further expands the application portfolio of high aspect ratio TiO₂ nanotubes that were already successfully utilized for water-splitting, photocatalysis, dye-sensitized solar cells, etc.^{61–63}

CONCLUSION

In conclusion, an optimization of the ALD coating process of very high aspect-ratio (≈ 180) TiO₂ nanotubular structures by Al₂O₃ was successfully carried out. The influence of the diffusion time of the TMA precursor on the Al₂O₃ coating was studied during the deposition processes. SEM inspection verified that short TMA exposure times (0.5 and 2 s) led to inhomogeneous coating. In contrast, a homogeneous coating of the nanotube interior was achieved by longer TMA exposure times (5 and 10 s). The QCM results pointed out on the clogging of the intertube space at an early stage of the coating process. This clogging was confirmed by detailed SEM analysis. The intertube clogging led to a decrease of the available area and hence to a decrease of the mass uptake, as reflected by the changes within the QCM measurements. Additionally, theoretical exposure minimum time for a complete and uniform coating within the TiO₂ nanotubular layers was calculated. Theoretical and experimental results clearly matched the diffusion-limited regime at the deposition process, and helped to gain important information for the application of the optimum deposition conditions, in high aspect ratio nanotubular structure. Finally, uniform, continuous, and pinhole-free Al₂O₃ coating on the inner wall of the TiO₂ nanotubes were confirmed by STEM images. These promising results motivate us to further optimize the ALD process toward coatings or complete inner fillings of even higher aspect ratio nanotube layers with various secondary materials. Such composite nanotube layers could open a promising pathway for further exploration of the exceptional inherent properties of self-organized nanotube TiO₂ layers.

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Notes

The authors declare no competing financial interest.

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